Homework 1: Statistical Review

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Problem 1

A melting point test of `n = 10` samples of a binder used in manufacturing a rocket propellant resulted in $\bar{x} = 154.2°F$. Assume that the melting point is normally distributed with $\sigma = 1.5°F$.

1. Test $H_0 : \mu = 155$ versus $H_1 : \mu \neq 155$ using $\alpha = 0.01$. 2. What is the p-value for this test? 3. What is the probability of committing a Type II error, β, if the true mean is $\mu = 150$?

In your submitted response, include the hypotheses, test statistic, rejection rule or p-value decision, and conclusion. For the Type II error part, explain the acceptance region under H_0 and evaluate its probability under $\mu = 150$.

Problem 2

The brightness of a television picture tube can be evaluated by measuring the amount of current required to achieve a particular brightness level. A sample of 10 tubes results in $\bar{x} = 317.2$ and $s = 15.7$.

Find a 99% confidence interval, in microamps, for the mean current required. State any necessary assumptions about the underlying distribution of the data.

In your submitted response, include the confidence interval formula, the critical value, the interval, and an interpretation in context.

Problem 3

Consider the hypothesis test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$.

Suppose that $n_1 = 15$, $n_2 = 15$, $\bar{x}_1 = 4.7$, $\bar{x}_2 = 7.8$, $s_1^2 = 4$, and $s_2^2 = 6.25$. Assume that $\sigma_1^2 = \sigma_2^2$ and that the data are drawn from normal distributions. Use $\alpha = 0.05$.

1. Test the hypothesis. 2. Find the p-value.

In your submitted response, include the pooled variance, degrees of freedom, test statistic, p-value, and conclusion in context.

Problem 4

Two chemical companies can supply a raw material. The concentration of a particular element in this material is important. The mean concentration for both suppliers is the same, but you suspect that the variability in concentration may differ for the two companies.

The standard deviation of concentration in a random sample of $n_1 = 10$ batches produced by company 1 is $s_1 = 4.7$ grams per liter. For company 2, a random sample of $n_2 = 16$ batches yields $s_2 = 5.8$ grams per liter.

Is there sufficient evidence to conclude that the two population variances differ? Use $\alpha = 0.05$.

In your submitted response, include the hypotheses for a two-sided variance comparison, define your F statistic, give the p-value or critical-value decision, and state the conclusion in context.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: “No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.